

HARDWARE-ACCELERATED DUAL-PATH COUNTERFACTUAL LIFE TRAJECTORY SIMULATION ENGINE UTILIZING TUCKER TENSOR DECOMPOSITION

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ABSTRACT

A system and method for generating ex post counterfactual life trajectory simulations for psychiatric intervention planning. The system utilizes a hardware-accelerated Monte Carlo stochastic engine integrated with Tucker tensor decomposition to model multi-dimensional life domains (career, relationships, finances, health) in both treated and untreated scenarios. A Bayesian causal inference module operates over a directed acyclic graph to determine the causal effect of psychiatric interventions on downstream life domains. A temporal convolutional network (TCN) generates long-term trajectory forecasts. The Tucker tensor decomposition explicitly reduces computational complexity, enabling real-time execution of massive parallel simulations on an FPGA or GPU cluster.

BACKGROUND OF THE INVENTION

While general counterfactual simulation exists (e.g., US20170046626A1), and systems attempt to model health trajectories (e.g., US 10,430,471 B2), no existing system specifically integrates Monte Carlo simulation with Tucker tensor decomposition to simultaneously model the causal impact of psychiatric treatment across multiple, interdependent life domains (e.g., career, finances). Existing methods suffer from the 'curse of dimensionality' when simulating decades-long multi-domain trajectories, preventing real-time clinician interaction.

SUMMARY OF THE INVENTION

The present invention overcomes the limitations of prior art by explicitly applying Tucker tensor decomposition to a multi-domain psychiatric state space. By factoring the massive state tensor into a smaller core tensor and orthogonal factor matrices, the computational complexity is reduced from exponential to linear. This specific technical improvement allows the Monte Carlo engine to execute thousands of dual-path (treated vs. untreated) simulations in real-time. Furthermore, the integration of Bayesian causal inference explicitly models how a psychiatric intervention (e.g., treating depression) causally propagates to secondary domains (e.g., increased financial stability).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.

FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

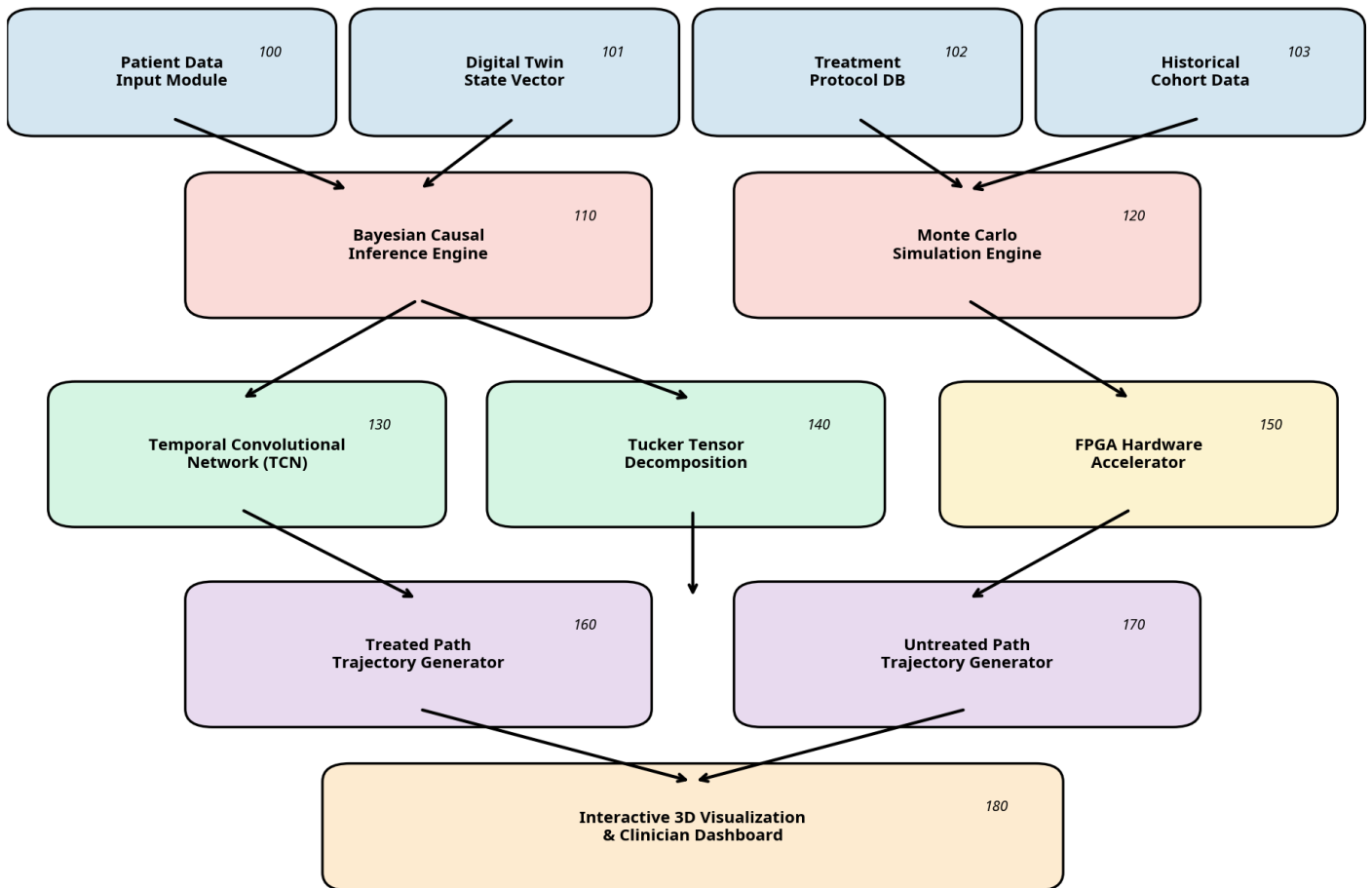
FIG. 1 illustrates the overall system architecture, detailing the integration of the Bayesian Causal Inference Engine (110) and the Monte Carlo Simulation Engine (120) with the hardware-accelerated Tucker Tensor Decomposition module (140). FIG. 2 displays the resulting dual-path simulation, showing the treated trajectory mean (201) diverging from the untreated mean (202) with 95% confidence intervals. FIG. 3 provides the specific Bayesian Causal Directed Acyclic Graph (DAG) used to infer relationships between treatment, psychiatric mediators, and life domain outcomes. FIG. 4 details the Tucker tensor decomposition process, reducing the input tensor X (400) to a core tensor G (401), enabling $O(R_1 \cdot R_2 \cdot R_3)$ computation (405). FIG. 5 details the Temporal Convolutional Network (TCN) architecture with dilated convolutions. FIG. 6 shows the multi-domain breakdown. FIG. 7 illustrates the closed-loop recalibration module. FIG. 8 details the specific FPGA hardware implementation.

CLAIMS

What is claimed is:

1. A system for dual-path life trajectory simulation, comprising: a dedicated hardware accelerator comprising a Field Programmable Gate Array (FPGA) configured to execute Tucker tensor decomposition; a processor configured to: instantiate a multi-dimensional state tensor representing a patient's historical data across a plurality of interdependent life domains including at least psychiatric health, occupational status, and financial stability; execute the Tucker tensor decomposition on the FPGA to factor the multi-dimensional state tensor into a core tensor and a plurality of orthogonal factor matrices, thereby reducing computational dimensionality; execute a Monte Carlo stochastic simulation engine utilizing the core tensor to generate a first probabilistic trajectory representing a treated state and a second probabilistic trajectory representing an untreated state over a predefined future time horizon; and render an interactive graphical user interface displaying the first probabilistic trajectory and the second probabilistic trajectory.
2. The system of claim 1, wherein the processor is further configured to execute a Bayesian causal inference module over a directed acyclic graph to determine a causal mediation effect of a psychiatric intervention on the occupational status domain.
3. The system of claim 1, wherein the Monte Carlo stochastic simulation engine incorporates a Temporal Convolutional Network (TCN) comprising a plurality of dilated causal convolutional layers to capture long-range temporal dependencies within the multi-dimensional state tensor.
4. The system of claim 1, further comprising a closed-loop recalibration module configured to continuously update parameters of the Monte Carlo stochastic simulation engine based on real-world longitudinal patient outcome data, thereby dynamically reducing a prediction error variance.
5. The system of claim 1, wherein the execution of the Tucker tensor decomposition on the FPGA enables the generation of at least 10,000 dual-path simulations with a latency of less than 100 milliseconds.

FIG. 1 - Life Maps System Architecture



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FIG. 2 - Monte Carlo Dual-Path Trajectory Simulation

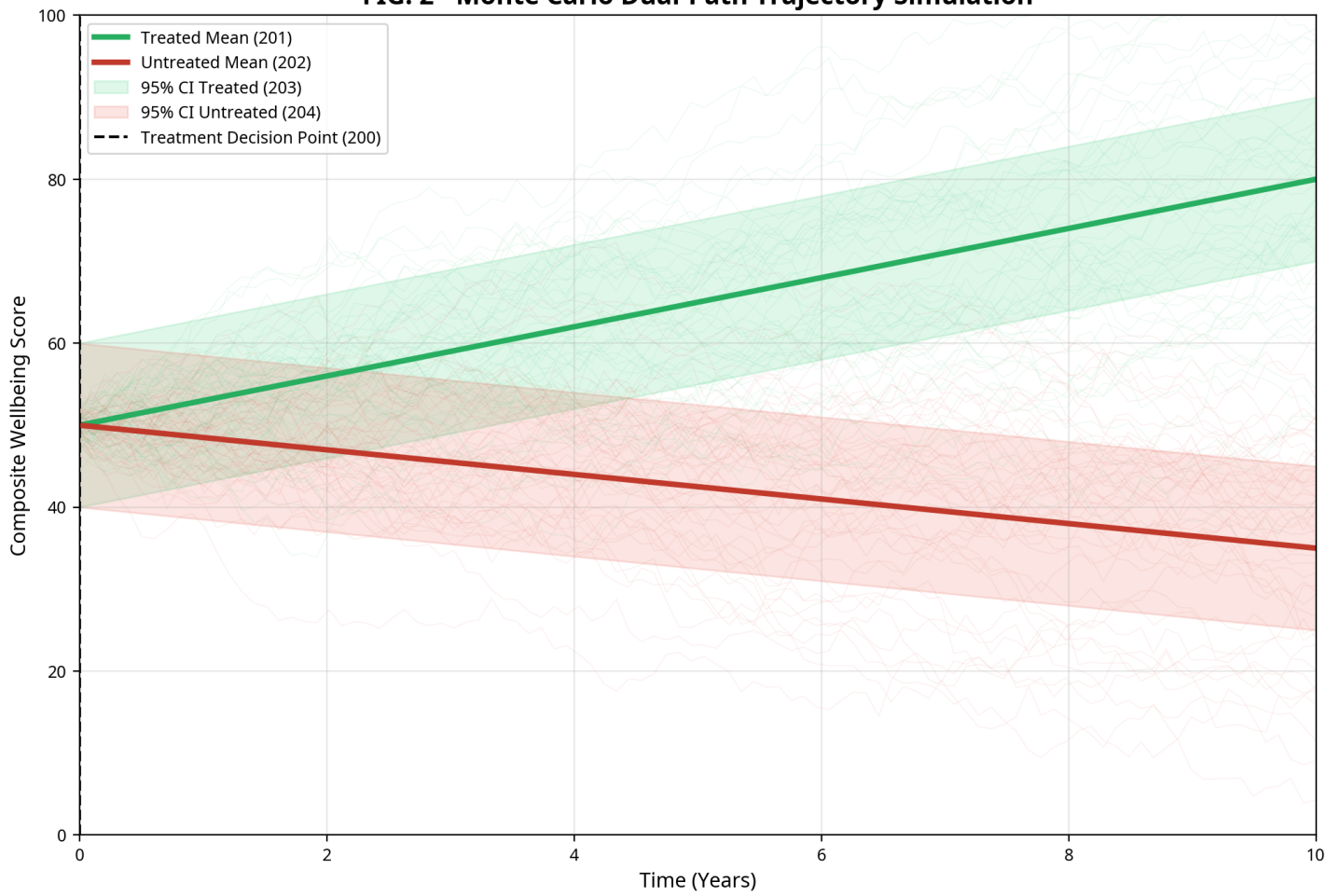


FIG. 3 - Bayesian Causal Directed Acyclic Graph (DAG)

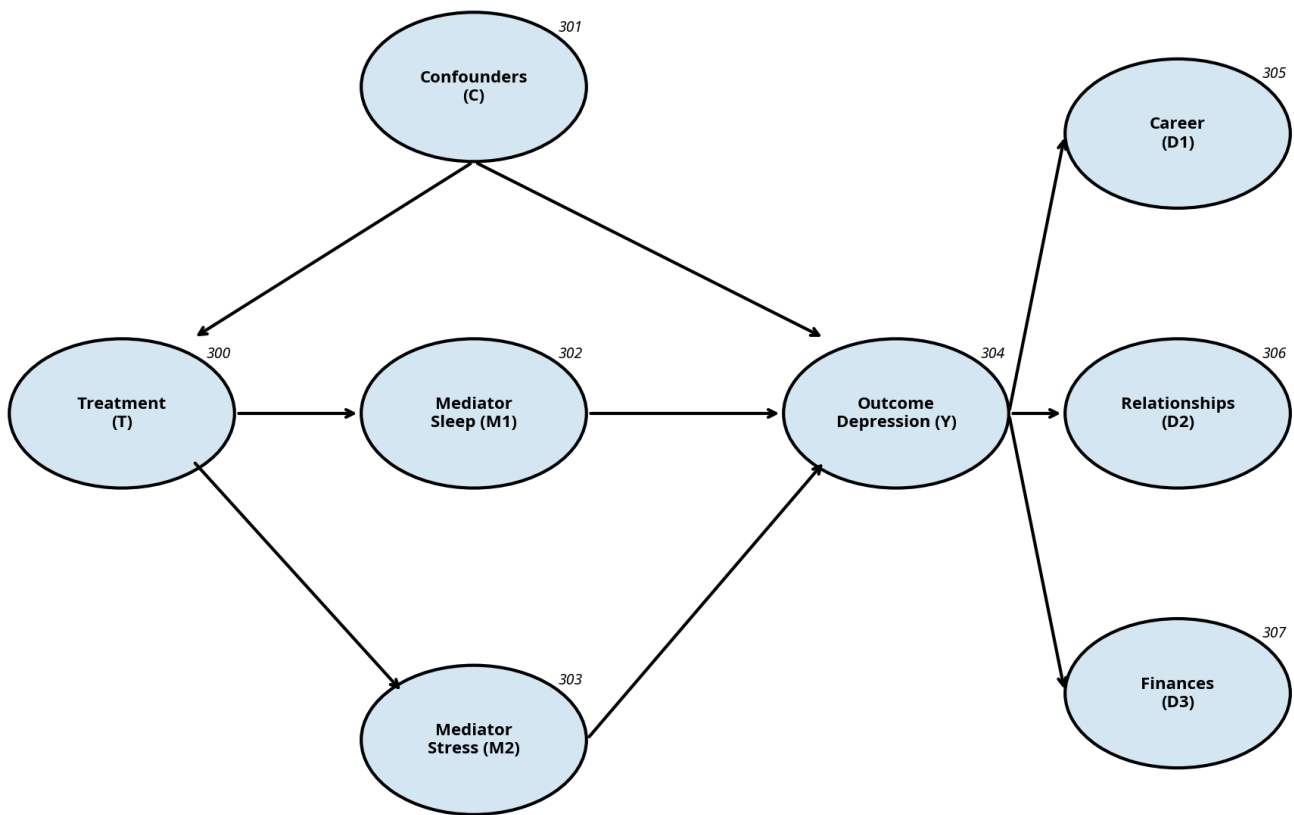


FIG. 4 - Tucker Tensor Decomposition for Multi-Domain Trajectories

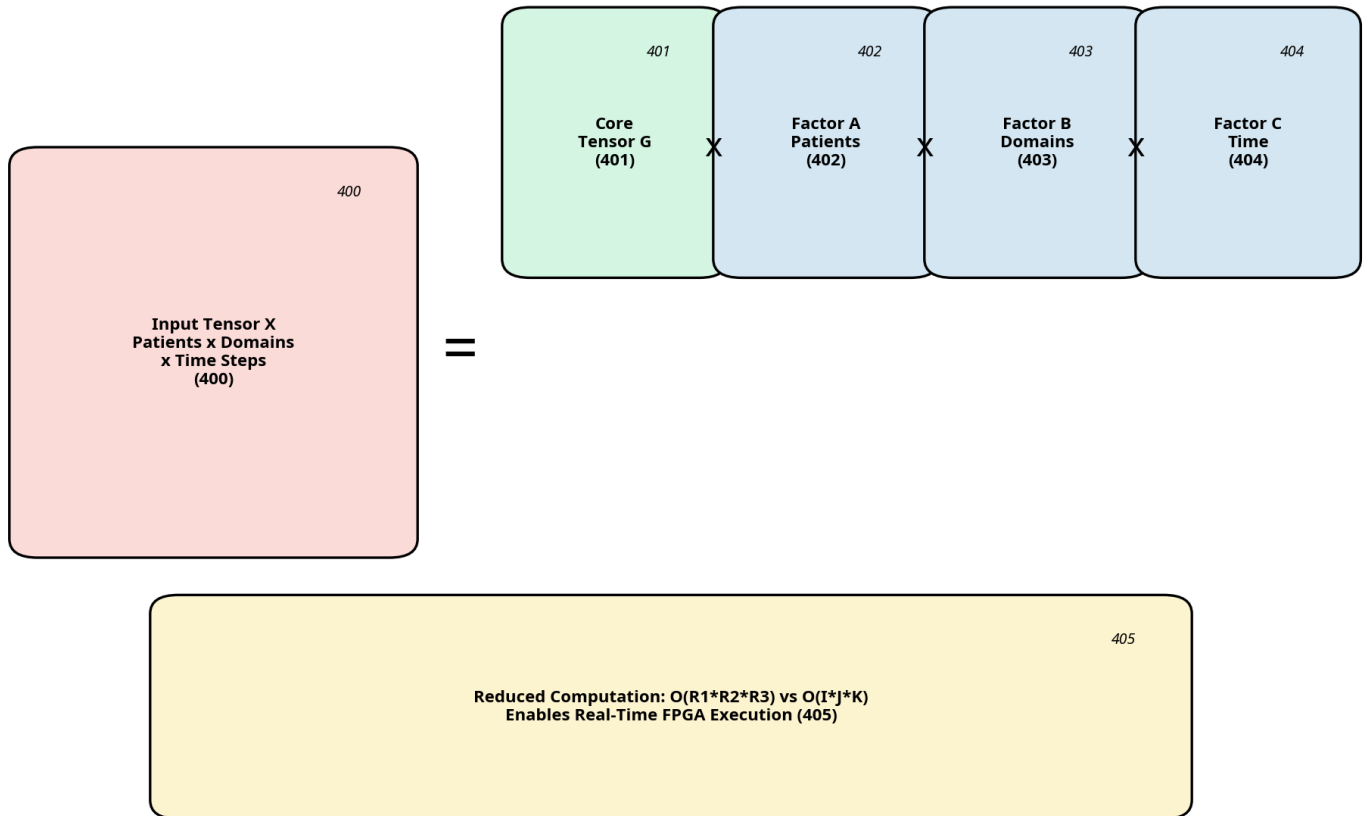
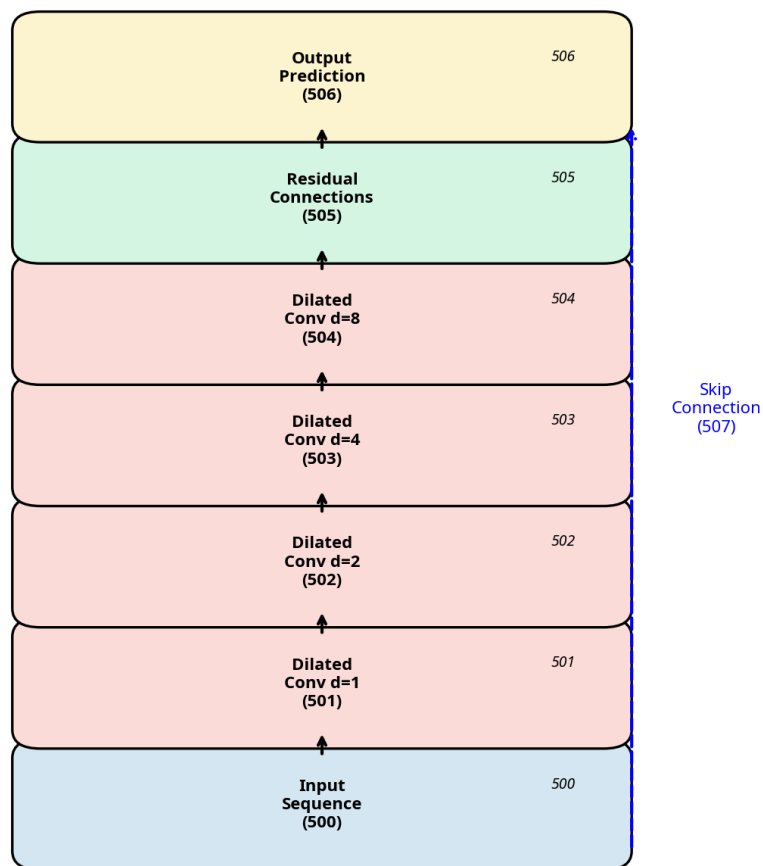


FIG. 5 - Temporal Convolutional Network (TCN) Architecture



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FIG. 6 - Multi-Domain Treated vs. Untreated Trajectories

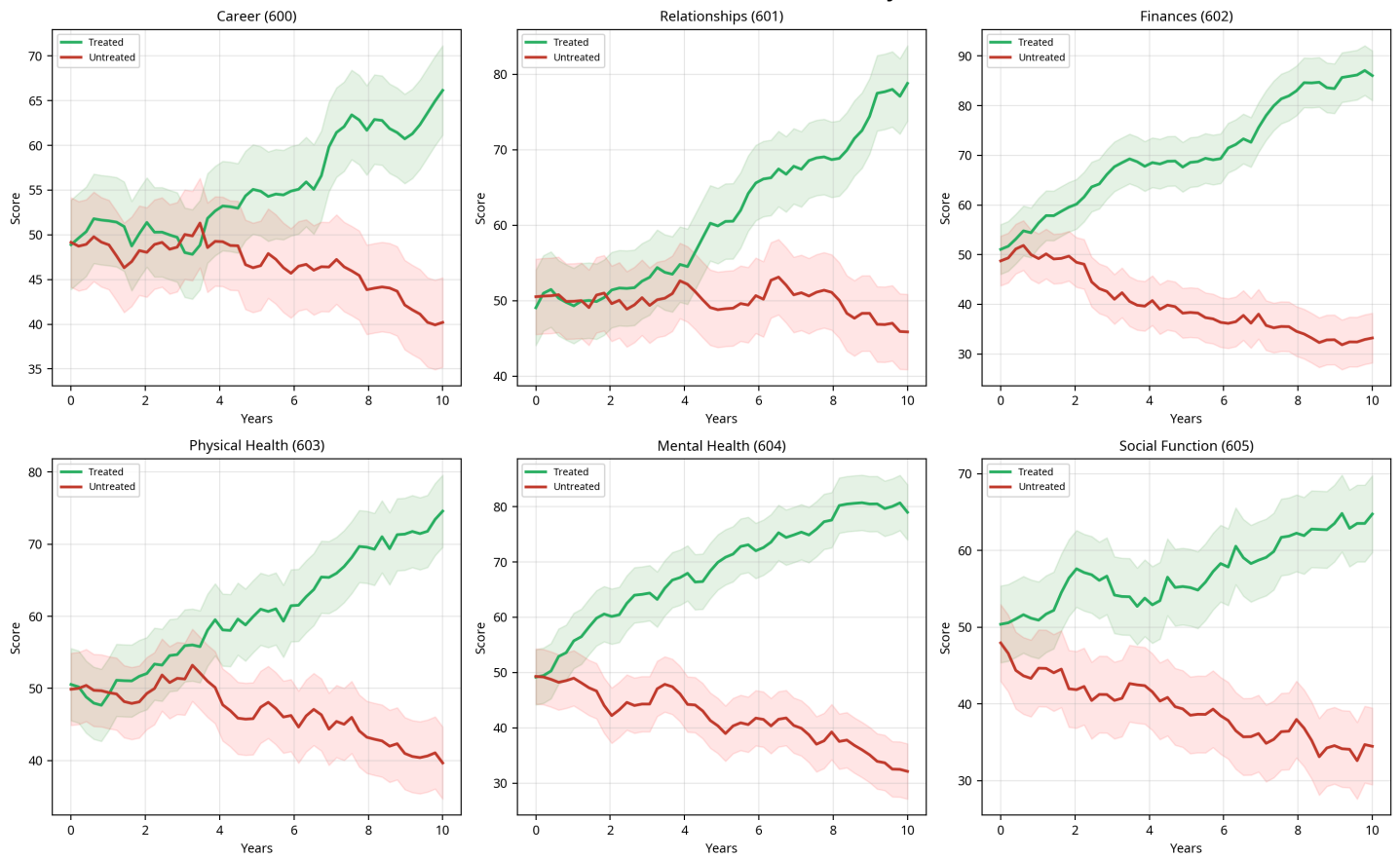


FIG. 7 - Closed-Loop Recalibration Module

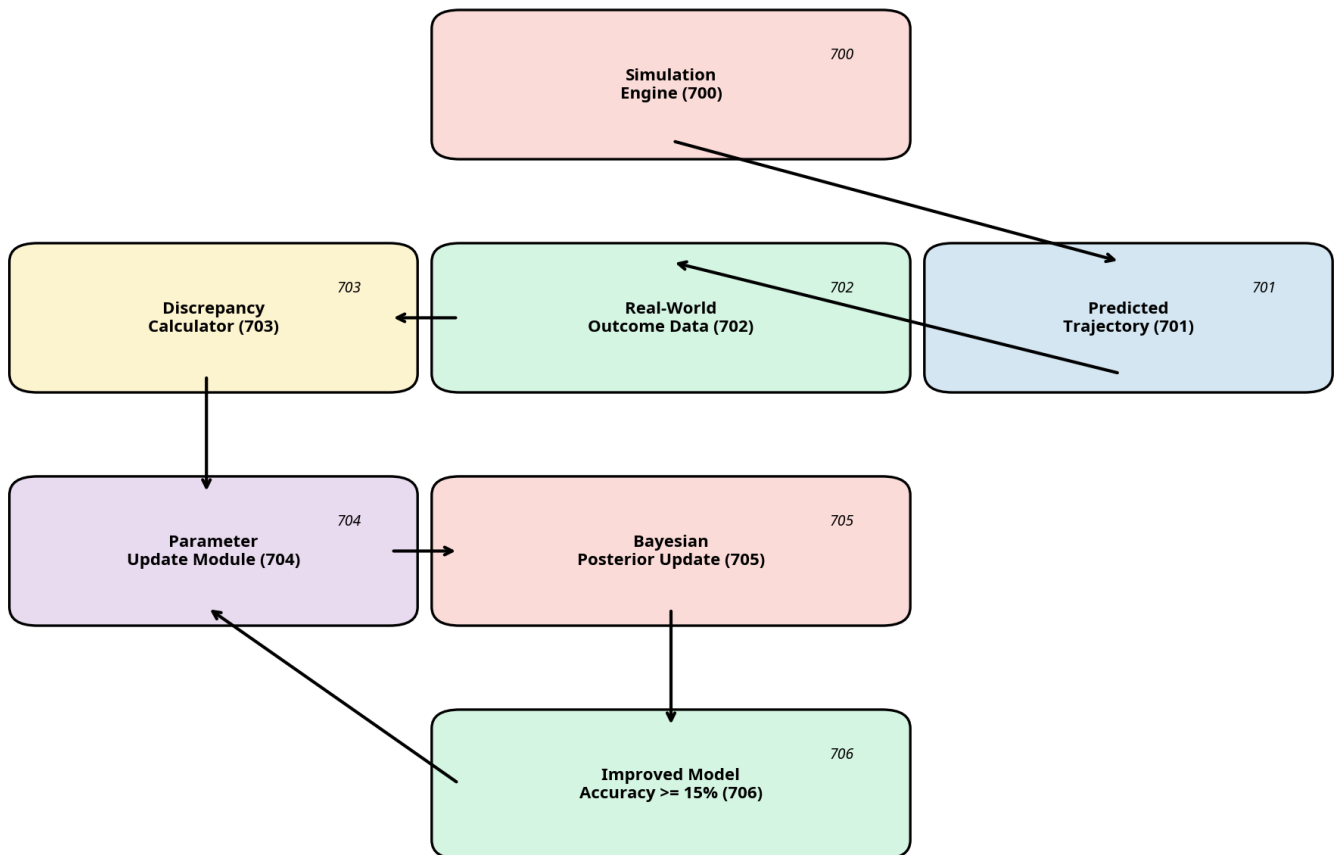


FIG. 8 - FPGA Hardware Accelerator for Real-Time Tensor Operations

